Generative AI · CA Mini-Project Report · Track D (Generation tools)

P39 · Class-Conditional Image Studio (BigGAN)

Free text → nearest ImageNet class (LLM) → BigGAN-deep-128 → CLIP ranking

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Class	AIA5
Course	Generative AI (Units I–V)
Problem	P39 · Class-Conditional Image Studio (BigGAN)
Live app	https://p39-biggan-studio-2yj4yrxzre63hg8dhuqdkt.streamlit.app/
Source code	https://github.com/shreyanshu1966/p39-biggan-studio

Abstract. A class-conditional GAN can only draw the 1000 ImageNet classes it was trained on, while people describe images in free text. This project bridges the two. A large language model (gpt-oss-20b) maps any description to the nearest ImageNet class, a pretrained **BigGAN-deep-128** generates samples with a user-controlled truncation ψ, and **CLIP** ranks the samples against the original text. Prompting the LLM few-shot with retrieved candidates and query expansion raised text→class accuracy from 47% (embedding search) to 93–97% over two runs on 30 labelled prompts, with 0% invalid classes vs 27% for zero-shot. Raising ψ from 0.2 to 1.0 increased sample diversity about 4×, and a one-step diffusion model (sd-turbo) followed the text better (CLIP 32.2 vs 28.5) but was about 18× slower on CPU. The app is deployed on Streamlit Community Cloud and includes a safety guardrail, a misuse note, a class-imbalance probe and a model card.

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1. Introduction

Allocation: Roll No. 39 → **P39 · Class-Conditional Image Studio (BigGAN)**, Track D (generation tools, so Unit IV is built deepest). **Goal:** generate images from free text via the nearest ImageNet class.

1.1 Problem statement and objectives

- Accept any free-text description and choose the most suitable of the 1000 ImageNet-1k classes.
- Generate class-conditional images with a pretrained BigGAN and expose the truncation trick as a control.
- Rank the generated samples with CLIP and compare BigGAN against a text-to-image diffusion model.
- Measure mapping accuracy, hallucination, diversity and speed, and document at least one failure case.
- Ship it as a public web app with a guardrail, a misuse note, licence notes and a model card.

1.2 Deliverables

Table 1. *Project deliverables.*

Deliverable	Where
Live studio app	https://p39-biggan-studio-2yj4yrxzre63hg8dhuqdkt.streamlit.app/
Public GitHub repository (README, model card, evaluation scripts)	https://github.com/shreyanshu1966/p39-biggan-studio
5 prompts × 3 truncation values grid	Figure 6, results/grid_5x3.png
Measured results and failure cases	Section 4, Section 5 and Appendix A
2–3 minute demo video	P39_Demo_Video.mp4 (submitted with this report)

2. System design

Figure 1 shows the project's generative-AI lifecycle. At run time a request passes through five stages: an input guardrail; text→class mapping (the LLM first guesses class names, the MiniLM sentence encoder retrieves candidate labels for the text and the guesses, and the LLM picks one by index); a second guardrail check on the chosen class; generation by BigGAN with truncation ψ; and CLIP ranking against the original text.

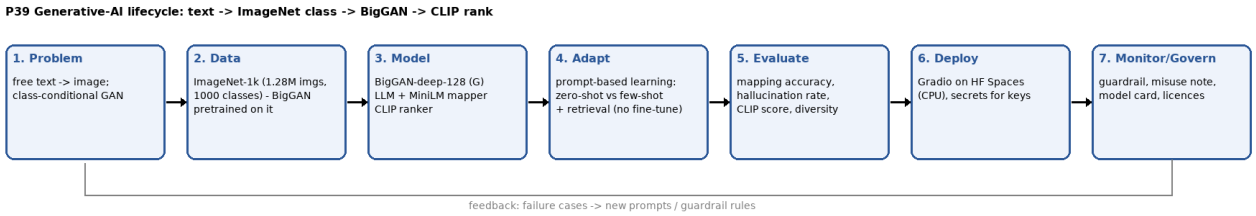


Figure 1. *Generative-AI lifecycle of the project, from problem framing to governance.*

Table 2. *Components of the pipeline.*

Component	Model / tool	Role	Licence
Generator	BigGAN-deep-128 (pytorch-pretrained-biggan)	Class-conditional 128×128 synthesis	MIT code, Apache-2.0 weights
LLM (GPT-style)	openai/gpt-oss-20b on Groq	Query expansion and few-shot class choice	Apache-2.0
Encoder (BERT family)	sentence-transformers/all-MiniLM -L6-v2	Candidate retrieval, baseline, confidence	Apache-2.0
Ranker (multimodal)	openai/clip-vit-base-patch32	Image–text CLIP score	MIT
Comparison only	stabilityai/sd-turbo	Diffusion baseline (U3)	Non-commercial research
App and hosting	Streamlit (Community Cloud), Gradio for local use	UI and deployment	Apache-2.0

3. Implementation by unit

3.1 Unit I · Foundations: truncation as a narrower distribution

Task from the problem statement: Explain truncation as sampling from a narrower distribution.

BigGAN's generator G(z, y) maps a latent vector $z \in R^{128}$ and a one-hot class y to an image; during training $z \sim N(0, I)$. The **truncation trick** draws z from a normal distribution truncated to $[-2, 2]$ and scales it by $\psi \leq 1$. The samples therefore come from a narrower region around the mode of the prior, where the generator has seen the most data: images become more typical and cleaner but less varied. This is a fidelity–diversity (precision–recall) trade-off chosen at sampling time, with no retraining. **Generative vs discriminative:** BigGAN is generative (it samples $x \sim p(x \mid y)$), while CLIP and the sentence encoder are used discriminatively to score how well an image or label matches the text. **Prompt techniques compared:** zero-shot vs few-shot prompting (Section 3.2 and Table 4).

3.2 Unit II · GANs and LLMs: text → class, then BigGAN

Task from the problem statement: Map text to a class with the LLM; generate with pretrained BigGAN.

The LLM is adapted by **prompt-based learning**, not fine-tuning. Four mappers were compared: plain embedding search (nearest label to the text); a **zero-shot** prompt asking for a class name, which is validated against the 1000 labels so invented names count as hallucinations; **few-shot + retrieval (v1)**, with three worked examples and the top 20 labels from MiniLM, where the answer must be an index from that list; and **few-shot + query expansion (v2)**, where the LLM first guesses three class names and candidates are retrieved for each guess as well as for the text. Every v1 error happened because the right class was missing from the top 20 ("the king of the jungle" is not near "lion" in embedding space). v2 removes that bottleneck.

Attention and conditioning in BigGAN: the class enters every residual block through conditional batch normalisation, whose gain and bias are predicted from [z, class embedding]. A self-attention block at 64×64 resolution lets distant regions of the image (two ears, several masts) stay consistent.

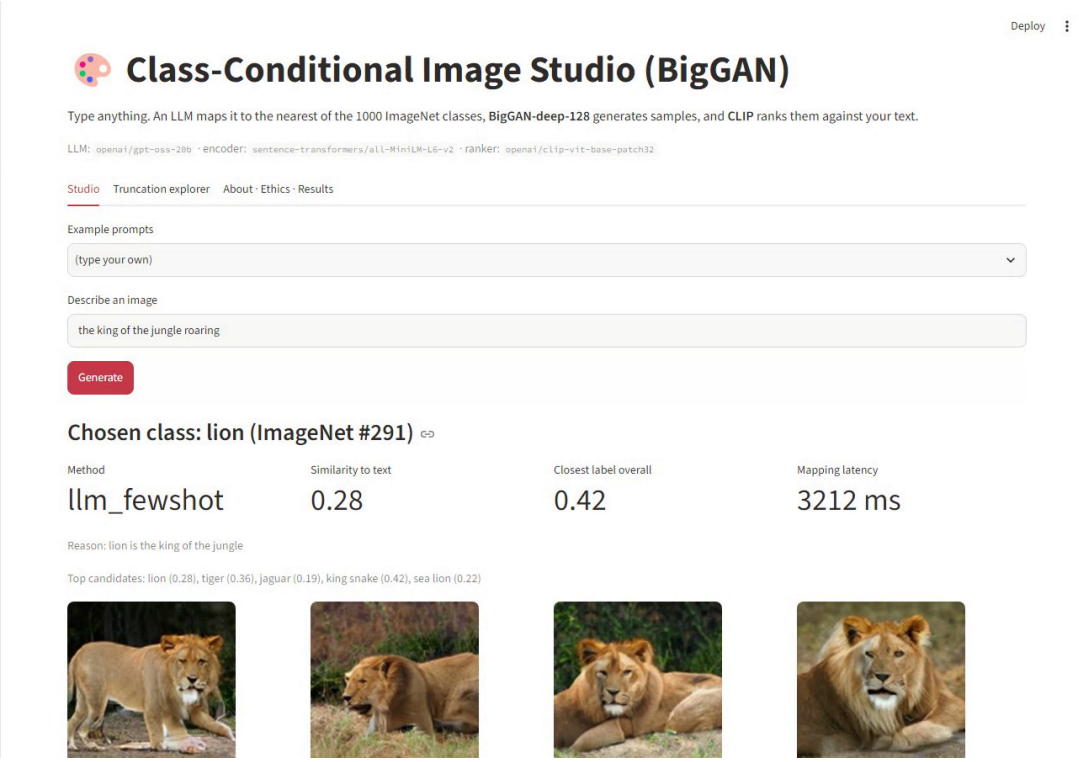


Figure 2. *Studio tab: "the king of the jungle roaring" is mapped to lion (#291) and four BigGAN samples are shown, ranked by CLIP score.*

3.3 Unit III · GPT, BERT and multimodal: CLIP ranking vs diffusion

Task from the problem statement: Rank outputs with CLIP; compare with a diffusion image.

Each sample is scored with CLIP ViT-B/32 as 100 × cosine(image embedding, text embedding), and the gallery is sorted by this score. The same five prompts were rendered with **sd-turbo**, a one-step text-to-image diffusion model, and both outputs were scored against the original free text (Table 6 and Figure 7). The pipeline therefore combines a GPT-style generator (gpt-oss-20b), a BERT-family encoder (MiniLM) and an image–text model (CLIP).

3.4 Unit IV · Tools: PyTorch truncation slider and grid (Track D focus)

Task from the problem statement: Build a truncation slider and grid in PyTorch.

Generation runs in PyTorch on CPU (about 2 s for four 128 px images). The app has a truncation slider (0.02–1.0), sample-count and seed controls, and a *Truncation explorer* that renders the same seed at ψ = 0.1, 0.3, 0.5, 0.7 and 1.0, so only the truncation changes (Figure 3). The required 5 prompts × 3 truncation values grid is Figure 6. The BigGAN weights are downloaded from the official pytorch-pretrained-biggan release and loaded with *torch.load(weights_only=True)*, so no pickled code is executed.

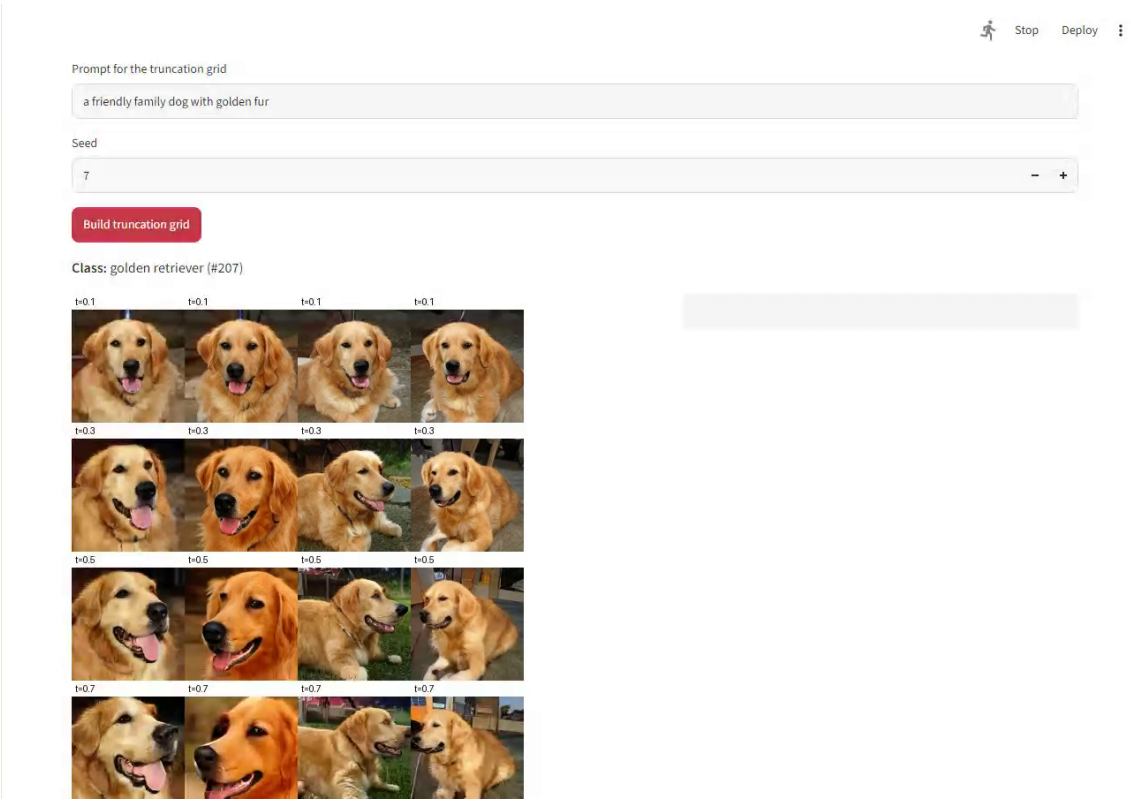


Figure 3. *Truncation explorer: the same seed at ψ = 0.1 → 0.7 (rows). Low ψ gives near-identical dogs; higher ψ gives more varied poses and backgrounds.*

3.5 Unit V · Responsible AI: misuse note and class imbalance

Task from the problem statement: Add a misuse note; discuss class imbalance.

- **Guardrail:** the input filter blocks violence, sexual content, hate, self-harm and real-person/deepfake requests (200-character limit). Weapon classes (assault rifle, revolver, rifle, missile …) are labelled, and blocked when combined with harmful-intent words (Figure 4).
- **Misuse note:** outputs are synthetic 128 px images of generic objects. They must not be presented as real photographs or used to deceive. The app is for education and research only.
- **Class imbalance:** ImageNet-1k has about 120 dog-breed classes but only three person-like classes (groom, scuba diver, ballplayer), and its photos skew towards Western web imagery. The people probe (Table A3) shows "a doctor" → lab coat, "a CEO in a meeting" → suit and "a farmer in India" → tractor.
- **Hallucination test:** zero-shot prompting produced invalid class names for 27% of prompts; the v2 mapper produced none, because every answer is checked against the label list.
- **Licences:** see Table 2. ImageNet and sd-turbo are non-commercial; a model card is in the repository.

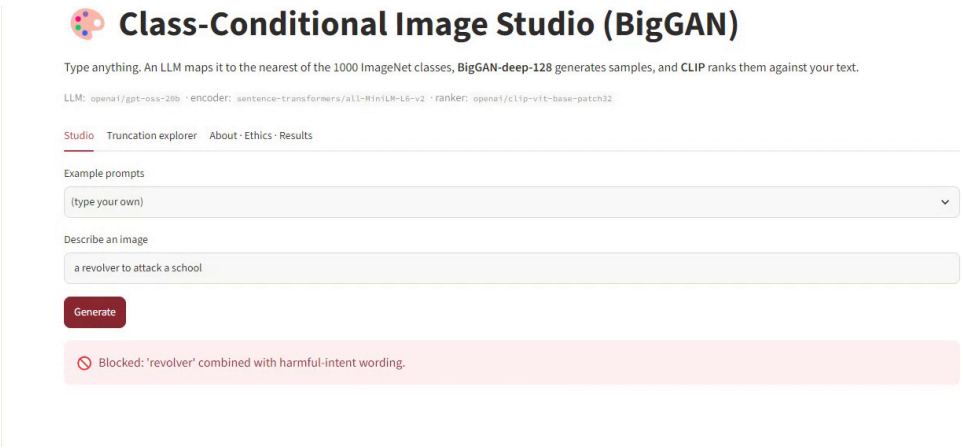


Figure 4. *Guardrail: "a revolver to attack a school" is blocked after class mapping.*

4. Results

4.1 Summary of measured results

Table 3. *Headline results.*

Measure	Result
Text → class accuracy (30 labelled prompts)	47% embedding → 90% zero-shot → 87% v1 → **93–97%** v2 (two runs)
Invalid (hallucinated) classes	zero-shot 27% · v2 0%
Truncation ψ 0.2 → 1.0	diversity 0.023 → 0.101 (≈4×); mean CLIP 28.4 → 28.0
BigGAN vs sd-turbo (mean CLIP)	28.5 vs 32.2
Generation speed (CPU)	BigGAN ≈ 2 s for 4 images · sd-turbo ≈ 36 s for 1 image at 512 px

4.2 Text → class mapping

Table 4. *Mapping accuracy on 30 hand-labelled prompts (latest run; v2 scored 29/30 in the previous run). Latency includes free-tier API calls; v2 makes two LLM calls.*

Method	Top-1 accuracy	Invalid classes	Mean latency
Embedding search (no LLM)	47% (14/30)	0%	16 ms
LLM zero-shot	90% (27/30)	27%	874 ms
LLM few-shot + retrieval (v1)	87% (26/30)	3%	3,144 ms
LLM few-shot + query expansion (v2)	93% (28/30)	0%	6,441 ms

The LLM methods are far more accurate than embedding search, which confuses surface words ("a stringed instrument played with a bow" → *bow*, "the king of the jungle" → *king snake*). Zero-shot is fast and accurate when its answer is valid, but 27% of its answers were not real ImageNet names ("piano" instead of "grand piano") and had to be snapped to the nearest label. v2 combines the LLM's world knowledge with a constrained answer space. Every error is listed in Table A1.

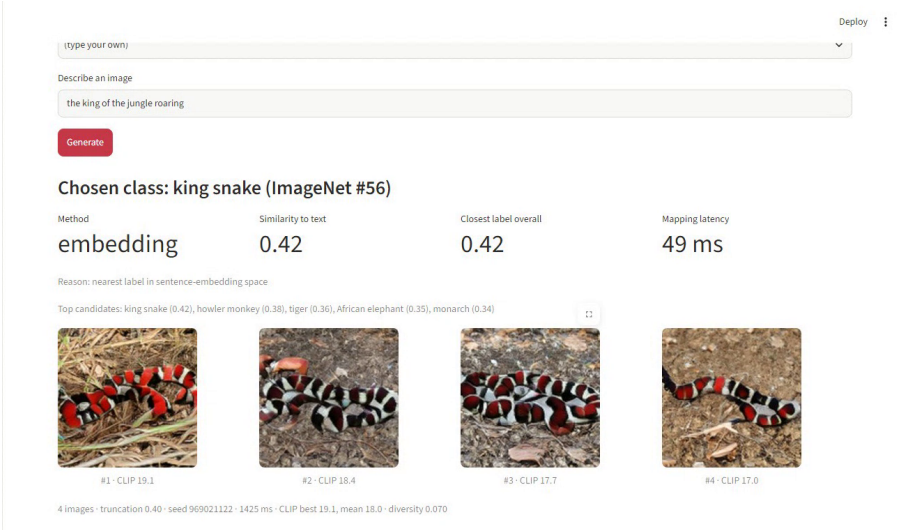


Figure 5. *Embedding-only baseline on the same prompt picks king snake, showing why the LLM is needed.*

4.3 Truncation trade-off and the 5 × 3 grid



Figure 6. *Required deliverable: 5 prompts × 3 truncation values (columns: prompt → class; rows: truncation ψ; each cell is the CLIP-best of 4 samples, same seed per prompt).*

Table 5. *Mean CLIP score and diversity (mean pairwise 1 − cosine of CLIP image embeddings) over 8 samples per cell.*

Prompt → class	CLIP ψ 0.2	CLIP ψ 0.5	CLIP ψ 1.0	Div. ψ 0.2	Div. ψ 0.5	Div. ψ 1.0
a friendly family dog with golden fur → *golden retriever*	29.6	29.4	29.1	0.025	0.049	0.084
a mountain erupting with lava → *volcano*	26.8	26.0	26.0	0.021	0.063	0.145
a slice of italian food with cheese and tomato → *pizza*	27.7	27.7	27.7	0.029	0.043	0.093
an old wooden ship with pirate flags → *schooner*	26.5	26.8	26.4	0.028	0.045	0.108
an orange and black butterfly that migrates → *monarch*	31.2	30.6	30.6	0.013	0.026	0.076
Mean	28.4	28.1	28.0	0.023	0.045	0.101

Diversity rises steadily with ψ for every prompt (≈4× on average), while the CLIP score changes by less than one point. Truncation mostly controls variety, not relevance; relevance is decided by the class choice.

4.4 BigGAN vs diffusion

Table 6. *CLIP score against the free-text prompt (BigGAN ψ = 0.4, best of 4). *First call includes loading CLIP.*

Prompt	BigGAN class	BigGAN CLIP	sd-turbo CLIP	BigGAN time	sd-turbo time
a friendly family dog with golden fur	*golden retriever*	29.7	30.1	8.6 s*	35.2 s
a mountain erupting with lava	*volcano*	28.5	30.7	1.8 s	35.5 s
a slice of italian food with cheese and tomato	*pizza*	28.5	30.6	1.8 s	37.9 s
an orange and black butterfly that migrates	*monarch*	31.6	33.4	2.0 s	36.1 s
a cat riding a bicycle	*bicycle-built-for-two*	24.3	36.1	1.9 s	35.1 s
Mean		28.5	32.2		



Figure 7. *Same prompts rendered by BigGAN (top row, CLIP-best of 4) and sd-turbo (bottom row); CLIP scores are printed on each image.*

Diffusion follows the whole sentence (colours, actions, scenes) and wins on every prompt, by the widest margin on the compositional prompt "a cat riding a bicycle" (24.3 vs 36.1). BigGAN is limited to one class at 128 px but is about 18× faster on CPU and gives direct, interpretable control through the class and ψ.

5. Failure cases and limitations

- **Composition is lost:** "a cat riding a bicycle" → *bicycle-built-for-two*; one class cannot express two objects and an action (Figure 7, last column).
- **Attribute reasoning:** v2 maps "a sour green apple" → *lemon* instead of *Granny Smith*; in one run "a stringed instrument played with a bow under the chin" → *cello*.
- **Out-of-vocabulary concepts:** "a giraffe at sunset" (giraffe is not an ImageNet-1k class) and "the Eiffel Tower at night" (→ *water tower*) are forced onto the nearest class (Table A2).
- **Abstract prompts:** "freedom" has no close label (best similarity 0.23), so the app shows a low-confidence warning (Figure 8).
- **Non-determinism:** the hosted LLM varied by one prompt between runs even at temperature 0.
- **General limits:** 128 px output, 1000 fixed classes, and CLIP's own web-data biases in the ranking score.

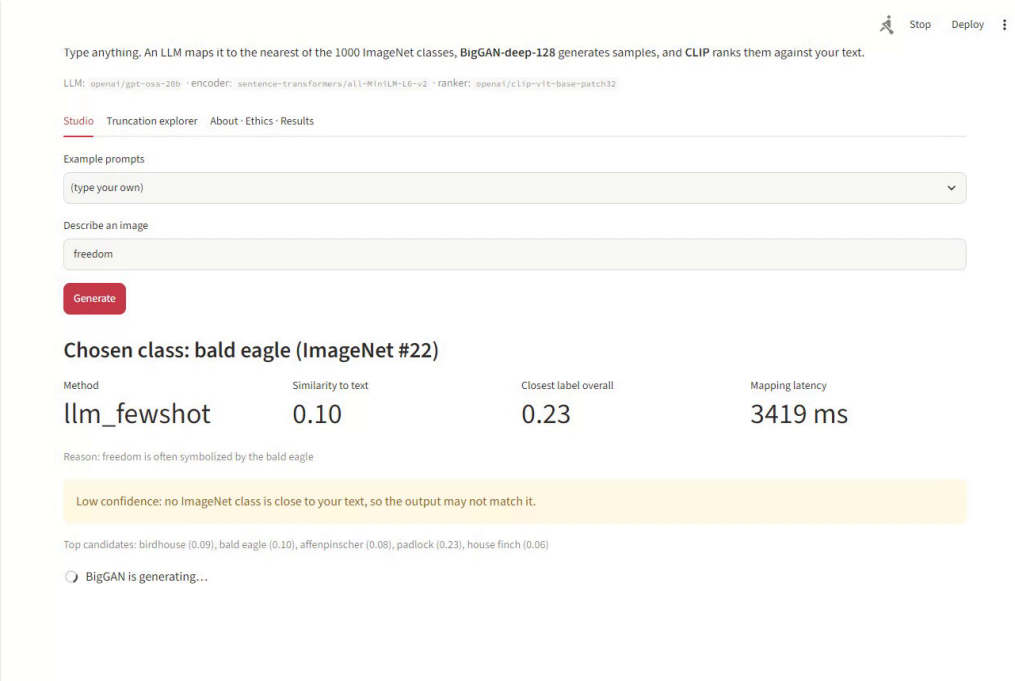


Figure 8. *Abstract prompt "freedom": no ImageNet class is close, so the app warns that the output may not match the text.*

6. Deployment and reproducibility

- **Live app:** https://p39-biggan-studio-2yj4yrxzre63hg8dhuqdkt.streamlit.app/ (Streamlit Community Cloud, file *streamlit_app.py*). The Groq API key is stored in the platform's secrets, never in the repository.
- **Code:** https://github.com/shreyanshu1966/p39-biggan-studio. *app.py* is the same app in Gradio for local use.
- **Reproduce the results:** *eval/run_mapping_eval.py*, *eval/make_grid.py* and *eval/compare_diffusion.py* regenerate every table and figure in this report; *docs/build_report.py* rebuilds this PDF.
- **Model card:** MODEL_CARD.md lists intended use, data, evaluation, biases and licences.

7. Conclusion and future work

The studio shows that an LLM is a practical bridge between open-ended language and a closed-vocabulary generator. Prompt engineering alone (few-shot examples, retrieval and query expansion, with no fine-tuning) took class accuracy from 47% to 93–97% and removed hallucinated classes. The truncation slider makes the GAN's fidelity–diversity trade-off visible and measurable. The diffusion comparison marks the limit of the approach: class-conditional GANs are fast and controllable but cannot express attributes or compositions.

Future work: blend two class embeddings for compositional prompts; let the LLM pick ψ from the wording ("a typical …" vs "unusual …"); upscale outputs with a super-resolution model; and replace the query-expansion call with a small fine-tuned classifier to cut latency.

References

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- O. Russakovsky et al. *ImageNet Large Scale Visual Recognition Challenge*. IJCV 2015.
- T. Brown et al. *Language Models are Few-Shot Learners*. NeurIPS 2020.
- OpenAI. *gpt-oss-20b model card*, 2025 (served via Groq).
- Hugging Face: pytorch-pretrained-BigGAN, transformers, sentence-transformers and diffusers libraries.

Appendix A. Detailed mapping results

Table A1. *Every wrong prediction on the 30 labelled prompts (latest run).*

Method	Prompt	Expected	Predicted
Embedding search (no LLM)	a fluffy orange cat sleeping	tabby, tiger cat, Persian cat, Egyptian cat	*Siamese cat*
Embedding search (no LLM)	the king of the jungle roaring	lion	*king snake*
Embedding search (no LLM)	a black and white bear eating bamboo	giant panda	*American black bear*
Embedding search (no LLM)	a big striped cat in the forest	tiger, tiger cat	*Madagascar cat*
Embedding search (no LLM)	a slow reptile with a shell walking on land	box turtle, mud turtle, terrapin	*alligator lizard*
Embedding search (no LLM)	a bird that cannot fly and lives in Antarctica	king penguin	*hummingbird*
Embedding search (no LLM)	the most feared predator of the ocean	great white shark, tiger shark	*sea snake*
Embedding search (no LLM)	the yellow vehicle that takes kids to class	school bus	*recreational vehicle*
Embedding search (no LLM)	a sour green apple	Granny Smith	*custard apple*
Embedding search (no LLM)	a small strong coffee in a tiny cup	espresso	*coffee mug*
Embedding search (no LLM)	a stringed instrument played with a bow under the chin	violin	*bow*
Embedding search (no LLM)	an instrument with black and white keys	grand piano, upright	*electric guitar*
Embedding search (no LLM)	a portable computer that folds open	laptop, notebook	*hand-held computer*
Embedding search (no LLM)	a tower with a light that guides ships	beacon	*water tower*
Embedding search (no LLM)	an orange and black butterfly that migrates	monarch	*sulphur butterfly*
Embedding search (no LLM)	an animal that hops and carries its baby in a pouch	wallaby	*beagle*
LLM zero-shot	a slow reptile with a shell walking on land	box turtle, mud turtle, terrapin	*leatherback turtle*
LLM zero-shot	a sour green apple	Granny Smith	*custard apple*
LLM zero-shot	an orange and black butterfly that migrates	monarch	*cabbage butterfly*
LLM few-shot + retrieval (v1)	the king of the jungle roaring	lion	*tiger*
LLM few-shot + retrieval (v1)	a sour green apple	Granny Smith	*orange*
LLM few-shot + retrieval (v1)	an orange and black butterfly that migrates	monarch	*sulphur butterfly*
LLM few-shot + retrieval (v1)	an animal that hops and carries its baby in a pouch	wallaby	*beagle*
LLM few-shot + query expansion (v2)	a sour green apple	Granny Smith	*lemon*
LLM few-shot + query expansion (v2)	a stringed instrument played with a bow under the chin	violin	*cello*

Table A2. *Out-of-vocabulary prompts: class chosen by each method.*

Prompt	Embedding	Zero-shot	Few-shot v1	Few-shot v2	Best sim.
a giraffe at sunset	polecat	hognose snake	polecat	hognose snake	0.504
a dragon breathing fire	European fire salamander	fire engine	Komodo dragon	Komodo dragon	0.57
freedom	padlock	padlock	prison	bald eagle	0.233 ⚠
a cat riding a bicycle	Egyptian cat	bicycle-built-for-two	tabby	bicycle-built-for-two	0.513
the Eiffel Tower at night	water tower	water tower	water tower	water tower	0.512
a smartphone with a cracked screen	screen	cellular telephone	cellular telephone	cellular telephone	0.551

Table A3. *People / bias probe: ImageNet has almost no person classes.*

Prompt	Embedding	Zero-shot	Few-shot v1	Few-shot v2	Best sim.
a doctor	medicine chest	medicine chest	lab coat	lab coat	0.467
a nurse	medicine chest	medicine chest	medicine chest	medicine chest	0.395
an engineer at work	power drill	carpenter's kit	power drill	laptop	0.418
a CEO in a meeting	entertainment center	harvestman	desk	suit	0.231 ⚠
a farmer in India	tractor	tractor	tractor	tractor	0.433
a bride	gown	gown	gown	gown	0.483

⚠ = low-confidence warning shown (best encoder similarity below 0.35).